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## **Bachelor of Computer Applications**

**Semester – V**

**Business Analytics**

**Experiment No.: 4**

**Title: Data Cleaning Using Power Query**

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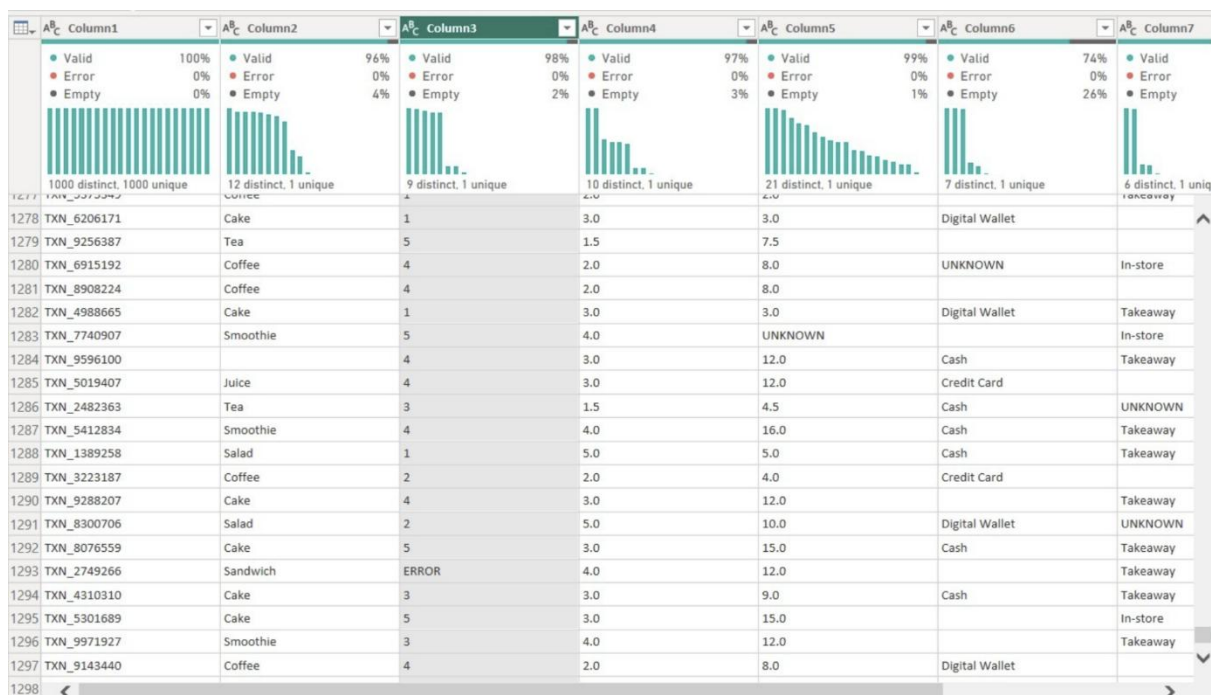
# Data Cleaning Using Power Query

## Problem Statement

A company has received customer sales data containing **missing values, duplicate records, invalid entries, and inconsistent data**. The objective is to clean and prepare the dataset using **Power Query** before performing further analysis.

### Step 1: Data Collection

- Obtained the **Cafe Sales Dirty Data** dataset from Kaggle.
- The dataset contains transaction information such as:
  - Transaction ID
  - Item
  - Quantity
  - Price Per Unit
  - Total Spent
  - Payment Method
  - Location
  - Transaction Date



### Step 2: Data Import

- Opened **Power BI Desktop**.
- Selected **Get Data** → **Text/CSV**.
- Imported the Cafe Sales CSV dataset.
- Selected **Transform Data** to open the dataset in **Power Query Editor**.

### Step 3: Data Cleaning and Transformation

- Used **Power Query Editor** to inspect the dataset.
- Used **Column Quality** to identify valid, empty, and error values.
- Removed **duplicate records** using Transaction ID.
- Identified and handled invalid values such as **ERROR** and **UNKNOWN**.
- Handled missing values by keeping them as blank/null where the correct value could not be determined.
- Cleaned inconsistent text entries in categorical columns.
- Corrected the data types of the columns.

### Data Types Applied

| Column           | Data Type      |
|------------------|----------------|
| Transaction ID   | Text           |
| Item             | Text           |
| Quantity         | Whole Number   |
| Price Per Unit   | Decimal Number |
| Total Spent      | Decimal Number |
| Payment Method   | Text           |
| Location         | Text           |
| Transaction Date | Date           |

The screenshot displays the Power Query Editor interface. The main area shows a table with the following columns: Transaction ID, Item, Quantity, Price Per Unit, and Total Spent. Each column has a data type dropdown and a 'Column Quality' icon. The 'Column Quality' icons show the distribution of data: Valid (green), Error (red), and Empty (grey). The 'Transaction ID' column is set to Text and has 1000 distinct values. The 'Item' column is set to Text and has 10 distinct values. The 'Quantity' column is set to Whole Number and has 6 distinct values. The 'Price Per Unit' column is set to Decimal Number and has 7 distinct values. The 'Total Spent' column is set to Decimal Number and has 19 distinct values. The 'Payment Method' column is set to Text and has 4 distinct values. The 'Location' column is set to Text and has 4 distinct values. The 'Transaction Date' column is set to Date and has 4 distinct values. The 'Query Settings' pane on the right shows the query name 'dirty\_cafe\_sales (1)' and a list of applied steps: Source, Changed column type, Removed Duplicates, Promoted Headers, Changed Type, Replaced Value, Replaced Value1, Replaced Value2, Replaced Value3, Replaced Value4, Replaced Value5, Replaced Value6, Replaced Value7, Replaced Value8, Replaced Value9, Replaced Value10, Replaced Value11, Replaced Value12, Replaced Value13, and Replaced Errors. The bottom of the screen shows a preview of the data with 19 rows and 5 columns.

## Step 4: Data Validation

- Checked the cleaned dataset using **Column Quality**.
- Verified that invalid errors were removed.
- Verified that numerical columns contained appropriate numeric data types.
- Verified that **Transaction Date** was converted into Date format.
- Checked the dataset for remaining missing values.
- Ensured that the data was ready for further analysis.

## Output

The final output is a **cleaned Cafe Sales dataset** prepared using Power Query.

The cleaned dataset contains:

- Unique transaction records
- Proper data types
- Cleaned invalid entries
- Handled missing values
- Consistent text values
- Properly formatted transaction dates

## Conclusion

The Cafe Sales dataset was successfully cleaned and transformed using **Power Query**.

Duplicate records, invalid values, missing values, inconsistent entries, and incorrect data types were identified and handled. The resulting dataset is structured and suitable for further data analysis.

## Learning Outcomes

Through this experiment, I learned:

- How to import CSV data into **Power BI**.
- How to use **Power Query Editor** for data cleaning.
- How to identify missing and invalid values using **Column Quality**.
- How to remove duplicate records.
- How to replace and handle ERROR and UNKNOWN values.
- How to handle missing values.
- How to change and verify appropriate data types.
- How to convert a text date column into **Date** format.
- How to validate cleaned data before analysis.
- How to prepare raw data for further **business analysis**.

GitHub link : [Kavivarshini13/Business-Analytics](https://github.com/Kavivarshini13/Business-Analytics)